

WHAT IS THIS QUESTION ASKING		
IF THE QUESTION SAYS OR SHOWS	IT IS	GO TO
a shell diagram, or asks for protons, neutrons, valence electrons	atomic structure	2 ATOMS
“electron configuration”, or an ion with a charge	configuration, and how ions differ	2 ATOMS
“molecular shape”, “bond angle”, VSEPR, a formula like BrF ₃	shape from electron pairs	2 ATOMS
“ionic or covalent”, “polar”, electronegativity	bonding and polarity	2 ATOMS
g/mol, “molar mass”, “how many moles”, “how many molecules”	the mole	3 MOLES
a gas with P, V, T, or “at STP”	gas laws	3 MOLES
mol/L, “concentration”, “diluted to”, a burette, a titration	solutions	3 MOLES
“equilibrium”, K _c , K _p , “shifts”, “Le Chatelier”	equilibrium	4 ACIDS
pH, pOH, [H ⁺], acid, base, K _a , buffer	the most common question in the paper	4 ACIDS, the pH tree
“given that...” then two or three equations with ΔH	Hess’s law	5 ENERGY
ΔH, ΔS, ΔG, “spontaneous”, “exothermic”	thermodynamics	5 ENERGY
“rate”, a collision picture, a Maxwell-Boltzmann curve, a catalyst	kinetics	6 ELECTRONS

WHAT IS THIS QUESTION ASKING		
IF THE QUESTION SAYS OR SHOWS	IT IS	GO TO
“oxidation number/state”, “oxidised”, “half-equation”	redox	6 ELECTRONS
two half-cells, a salt bridge, E° values, anode/cathode	electrochemistry	6 ELECTRONS

HOW TO USE THIS PACK

It is open book, so the textbook is not the problem — the clock is. This pack is not a summary. It is here so you never have to go looking for a method.

- Read the question and find its words in the table above.
- Flip to that numbered side — the strip along the top of every side tells you where you are.
- Find the row whose left column matches, and run the steps in the middle. The right column is the same thing worked through on a question from your own worksheets.

Anything in **red** is a place marks are commonly dropped, not just an emphasis.

If you cannot find it here, then it is a look-it-up question and the textbook is the right tool. Do those last.

NOT IN YOUR COURSE — DO NOT GO HUNTING FOR THESE		
TOPIC	WHAT IS ABSENT	IF IT SEEMS NEEDED
Moles and gases	STP and 22.4 L/mol Gay-Lussac the combined gas law	your molar volume is 24 dm ³ at 20°C. Two-state problems run PV = nRT twice
Composition	percent composition empirical and molecular formula	absent from A105 entirely
Concentration	ppm % w/v	your four units are % by mass, g/L, mol/kg, mol/L
Equilibrium	ICE tables K _p and partial pressures leaving pure solids out of K _c	every equilibrium you are given is homogeneous
Acids	K _w as a symbol K _a × K _b = K _w the “x is small” approximation pH of a salt	you are given pH + pOH = 14 as a bare fact. Weak acids always come with a % dissociation

NOT IN YOUR COURSE — DO NOT GO HUNTING FOR THESE		
TOPIC	WHAT IS ABSENT	IF IT SEEMS NEEDED
Energy	calorimetry, q = mcΔT	zero mentions in all 13 decks
Cells	cell notation Zn Zn ²⁺ Cu ²⁺ Cu the Nernst equation	everything is at standard conditions
Lesson 11	no deck and no worksheet exists	if it is examinable, the material is not in what you were given – ask

THE TRAPS THAT COST THE MOST, ALL SIX SIDES		
WHERE	THE TRAP	THE FIX
gases	Celsius into a gas law	K = °C + 273, every time
gases	dm ³ where m ³ is needed	R = 8.314 wants m ³ and Pa
titration	the mole ratio upside down	the bigger coefficient gets the bigger mole count
acids	stopping at pOH	pH = 14 – pOH
acids	the strong-acid method on a weak acid	a ≈ or a quoted K _a means weak
buffers	the ratio inverted	salt over acid. More salt → pH above pK _a
Hess	reversing without flipping the sign of ΔH	reverse flips the sign, scaling changes the size
Gibbs	kJ mixed with J	ΔS tables are in J. Divide by 1000
Gibbs	–T × (–ΔS) read as minus	two negatives make a plus
redox	the oxidising agent	it is the thing that gets reduced. The two words must be opposites
cells	multiplying E° when you scale a half-equation	never scale the volts
electrolysis	time in minutes or hours	Q = I × t needs seconds
everywhere	answering to the wrong precision	the energy worksheet asks for 2 decimal places. Show the working either way

VSEPR — SHAPE FROM THE ELECTRON PAIRS		
DOMAINS / LONE PAIRS	SHAPE (HER WORDING)	ANGLE, AND AN EXAMPLE
2 0 lone	Linear	180° CO ₂
3 0 lone	Trigonal planar	120° BF ₃
3 1 lone	Bent or angular	< 120° SO ₂ , O ₃
4 0 lone	Tetrahedral	109.5° CH ₄
4 1 lone	Trigonal pyramid	< 109.5°, about 107.5° NH ₃
4 2 lone	Bent or angular	much < 109.5°, about 104.5° H ₂ O
5 0 lone	Trigonal bipyramid	90° and 120° PCl ₅
5 1 lone	Seesaw (sawhorse)	< 90° and < 120° SF ₄
5 2 lone	T-shape	< 90° ClF ₃
5 3 lone	Linear	180° XeF ₂
6 0 lone	Octahedral	90° SF ₆
6 1 lone	Square pyramid	< 90° BrF ₅
6 2 lone	Square planar	exactly 90° XeF ₄
6 3 lone	T-shape	< 90°

GETTING TO THE SHAPE		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
“what is the molecular shape of...”	1. draw the Lewis structure 2. count electron domains on the central atom 3. count how many of those are lone pairs 4. read that cell of the table	CO ₂ : two double bonds → 2 domains → linear
what counts as one domain	a double or triple bond is ONE domain , same as a single	counting CO ₂ as 4 gives tetrahedral, which is wrong
a quick route for AX _n with H, F, Cl, Br or I	lone pairs = (valence electrons of A – n) ÷ 2, then domains = n + lone pairs	BrF ₅ : (7–5)/2 = 1 lone, 6 domains → square pyramid
“electron geometry” vs “molecular shape”	geometry is the 0-lone-pair name for that row. Shape is the actual cell	XeF ₄ is octahedral geometry but square planar shape
“will the angles differ?”	lone-lone > lone-bond > bond-bond repulsion. More lone pairs, smaller angle	1 lone → about 107.5° 2 lone → about 104.5° Different , and quote the numbers

GETTING TO THE SHAPE		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
a wedge-and-dash drawing	plain line in the plane, bold wedge forwards, dashed backwards	

ATOMS, IONS AND ISOTOPES		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
a shell diagram, or a box with blanks	1. protons = atomic number, the top number in the box 2. nuclear charge = + Z , write the plus sign 3. neutrons = mass number – Z 4. electrons = Z for a neutral atom only 5. valence electrons = the outermost ring 6. configuration fills 2, then 8, then 8	boron: 5p, 6n, mass 11, 2,3, charge 0
configuration of an ion	cation loses electrons, anion gains them. Protons and mass do not change	Mg ²⁺ : still 12p and 12n, but 10 electrons → 2,8
identify an atom from a diagram	count the + in the nucleus — that is the element. Then compare electrons to protons for the charge	more electrons than protons → anion
isotope abundances → relative atomic mass	RAM = Σ(isotope mass × its fraction)	Mg: 23.985(0.7899) + 24.986(0.1000) + 25.983(0.1101) = 24.31
counts given, not percentages	divide by the total first	329+829+529 = 1687, so 329/1687 = 19.50%
abundances unknown, RAM given	let x be the lighter fraction, solve one equation	10.013x + 11.009(1–x) = 10.881 → x = 12.88%
balance a nuclear equation	top numbers (mass) balance, bottom numbers (Z) balance, then read the element off Z. alpha is mass 4 / Z 2, beta is mass 0 / Z –1, positron is mass 0 / Z +1	U–238 → ? + alpha mass 238–4 = 234, Z 92–2 = 90 → Th–234
beta or positron?	your deck writes the positron with no plus sign	beta is –1 subscript, positron is +1. Misreading it changes the element

BONDING AND POLARITY		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
“what type of bond”	find both electronegativities, subtract, read the band	< 0.5 non-polar covalent 0.5–2.0 polar covalent > 2.0 ionic
worked, all three bands	H–F, K–Cl, Cl–Cl	1.78 polar covalent 2.34 ionic 0 non-polar
“is the MOLECULE polar”	1. are the bonds polar? (ΔEN) 2. what is the shape? 3. do the bond dipoles cancel?	polar bonds do not make a polar molecule. CO ₂ is linear, so they cancel → non-polar
symmetric AB ₄ with identical B	dipoles cancel → non-polar	change one B, or add a lone pair, and it becomes polar
“which has the highest boiling point”	strongest intermolecular force wins: ion-dipole > hydrogen bond > permanent dipole > temporary dipole	
does it hydrogen bond?	the H must be bonded TO the F, O or N — not merely present in the molecule	CH ₃ F contains F but every H is on carbon → no hydrogen bonding
electronegativity trend	increases across a period, decreases down a group	F is the most electronegative
atomic radius trend	decreases across a period, increases down a group	more shells → valence electrons further out and shielded

ATOMS — TWO THINGS YOUR NOTES DO NOT GIVE YOU	
There is no table of ion charges anywhere in your materials , and one worksheet question needs it. The periodic table you are given shows atomic number, symbol, name and relative atomic mass — no charges. Use the group rule: Gp 1 → 1+ Gp 2 → 2+ Gp 13 → 3+ Gp 15 → 3– Gp 16 → 2– Gp 17 → 1– Gp 18 → 0	
Then: metal + non-metal is ionic , two non-metals is covalent , and a covalent name tells you the count — trichloride is 3, tetrafluoride is 4. So magnesium chloride is MgCl₂ and aluminium oxide is Al₂O₃ .	
Metallic bonding is not in these two lessons at all — the only three types your deck offers are non-polar covalent, polar covalent and ionic.	
Your deck disagrees with itself twice. The VSEPR image says the tetrahedral angle is 109° while another slide says 109.5° — write 109.5°. And one slide calls relative atomic mass the “nucleon number”, which it is not: relative atomic mass is a weighted average across isotopes, which is why boron is 10.81 and not a whole number.	

UNITS AND CONSTANTS — CHECK THESE FIRST		
QUANTITY	UNIT HER COURSE USES	CONNECTS BY
amount	mol	$n = m \div M$
mass, molar mass	g, g/mol	$m = n \times M$
number of particles	atoms or molecules	$N = n \times 6.02 \times 10^{23}$
temperature	KELVIN, always	$K = ^\circ C + 273$ — her course uses 273, not 273.15
gas volume, gas pressure	m³ and Pa	because $R = 8.314 \text{ dm}^3 \rightarrow \text{m}^3$ is $\div 1000$, kPa $\rightarrow$ Pa is $\times 1000$
solution volume	L (= dm³)	$M = n \div V$
R , the gas constant	8.314 m³ Pa K⁻¹ mol⁻¹	the only one A105 uses. Every question quotes it
molar volume	24 dm³/mol at 20°C, 1 atm	your course has no STP and no 22.4. Recall only — every calculation goes through $PV = nRT$
volume chain	1 dm ³ = 1 L = 1000 cm ³ = 1000 mL = 0.001 m³	this one conversion is in every part of the gas question

MOLES, MASS AND PARTICLES		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
“molar mass of...”	1. count each atom, including subscripts outside brackets 2. add the relative atomic masses 3. unit is g/mol	CaCO ₃ = 40.08 + 12.01 + 3(16.00) = 100.09 Al ₂ (SO ₄) ₃ : the 3 multiplies S and all 4 0 $\rightarrow$ 342.17
grams given, moles wanted	$n = m \div M$	102 g HBr $\div$ 80.91 = 1.26 mol
moles given, grams wanted	$m = n \times M$	0.80 mol KBr $\times$ 119.00 = 95.20 g
“how many molecules / atoms”	$N = n \times 6.02 \times 10^{23}$	2.00 mol F ₂ $\rightarrow$ 1.20 $\times$ 10²⁴
“identify element X” from a mass and a mole count	1. get n of the compound from the mole ratio 2. $M = \text{mass} \div n$ 3. subtract the known atoms’ masses	$M(\text{XCl}_2) = 110.983$, minus 2(35.45) = 40.08 $\rightarrow$ Ca

GASES		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
Avogadro — moles change, T and P fixed	$V_1/n_1 = V_2/n_2$	5 mol in 50 L $\rightarrow$ 6 mol = 60 L. Only the TOTAL moles matter
Boyle — P and V, T fixed	$P_1V_1 = P_2V_2$	3 atm, 330 mL $\rightarrow$ 1 atm = 990 mL
Charles — V and T, P fixed	$V_1/T_1 = V_2/T_2$, T in K	25°C $\rightarrow$ 298 K is 298+343 K, Increase = 45/298 = 15.10%
anything else	PV = nRT	$n = PV/RT$ $V = nRT/P$ $T = PV/nR$
two states, both P and T change	your course has no combined gas law. Run PV = nRT twice, or take the ratio	$V_1/V_2 = (P_2/P_1)(T_2/T_1)$
“find the gas” from mass and PVT	1. $n = PV/RT$ 2. $M = \text{mass} \div n$ 3. match M to an element or compound	$M = 0.71(8.0100) = 71 \text{ g/mol} \rightarrow$ Cl₂
worked, all in base units	1 mol N ₂ , 0°C, 11.2 dm ³	$P = (1)(8.314)(273)/0.0112 =$ 202.65 kPa

SOLUTIONS, DILUTION AND TITRATION		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
“molarity”, mol/L, [X]	$M = n \div V$, V in litres	18.02 g glucose (M=180.18) in 250 mL $\rightarrow$ 0.100/0.250 = 0.400 M
the other three concentration units	% by mass = solute $\div$ solution g/L = mass $\div$ volume mol/kg = moles $\div$ solvent in kg	molality divides by the SOLVENT. The other two divide by the whole solution
“diluted to”, a volumetric flask	M₁V₁ = M₂V₂	V₁ is the FLASK’s capacity , not the water added. 25 mL into a 250 mL flask $\rightarrow$ $V_2 = 250$
dilution factor, serial dilution	$DF = V_{\text{final}} \div V_{\text{initial}}$	serial: multiply them. 10 $\times$ 20 = 200
titration — a burette table	1. throw away any trial marked Rough 2. volume used = final – initial, per trial 3. average the Accurate trials 4. $n = M \times V$ of the one you know 5. apply the mole ratio , then divide by the other volume	2NaOH + H ₂ SO ₄ : $n(\text{H}_2\text{SO}_4) = 3.78 \times 10^{-4}$ ratio 2:1 so $\times 2 \rightarrow 7.56 \times 10^{-4} \div 0.01000 \text{ L} =$ 0.0756 M

SOLUTIONS, DILUTION AND TITRATION		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
which way does the ratio go	the species with the bigger coefficient gets the bigger mole count	dividing by 2 in the example above gives 0.0189 and is wrong
pH given instead of a concentration	$[\text{H}^+] = 10^{-\text{pH}}$, then treat it as the acid’s molarity if monoprotic	pH 1.50 $\rightarrow$ $[\text{HCl}] = 0.0316 \text{ M}$

LIMITING REAGENT		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
“which is limiting”, or two amounts given	1. for EACH reactant, work out how much product it alone would make 2. the one giving the smaller product is limiting 3. the other is in excess	4 mol CO (1:1) $\rightarrow$ 4 mol H ₂ (2:1) $\rightarrow$ 3 H ₂ gives less $\rightarrow$ H₂ limiting
“how much is left over”	excess = starting amount – amount consumed	0.60 mol H ₂ , 0.40 consumed $\rightarrow$ 0.20 left
the trap	fewer starting moles does NOT mean limiting — the ratio decides	Zn 0.0642 vs CuSO ₄ 0.0500, ratio 1:1 $\rightarrow$ CuSO ₄ limits
one amount is a solution	$n = M \times V$ before comparing	0.250 M $\times$ 0.200 L = 0.0500 mol

THE LAB SESSION, AND WHAT YOUR COURSE LEAVES OUT	
Lesson 5 is the lab for Lesson 4 — standardising NaOH, and technique. No new arithmetic. The one written-answer fact: NaOH cannot be weighed accurately because it is hygroscopic and absorbs CO₂ , which is why it must be standardised against an acid of known concentration. Technique marks: read the meniscus at the bottom , no air bubble in the burette tip, add dropwise near the endpoint, phenolphthalein as indicator. Do not go looking for these — A105 does not teach them: STP and 22.4 L/mol; Gay-Lussac; the combined gas law; percent composition; empirical and molecular formula; ppm and % w/v. If a question seems to need one, re-read it. Deck artifact: the titration template slides print a stray “ $\times 2$ ” on an HCl + NaOH reaction that is 1 : 1 . The deck’s own worked answer uses no such factor. Use a 2 only when the balanced equation actually has one.	

"FIND THE PH OF..." — WHICH ONE IS IT		
WHAT YOU SEE	IT IS	WHAT TO DO
single arrow → in the dissociation, a named acid, no K given	strong acid	$[H^+] = (\text{number of } H^+) \times [\text{acid}]$, then $pH = -\log[H^+]$
NaOH , KOH, "alkaline", "base"	strong base	two steps, not one. $[OH^-] \rightarrow pOH = -\log[OH^-] \rightarrow pH = 14 - pOH$
double arrow ⇌, or a K_a is quoted	weak acid	you need the % dissociated – her course gives it
a % dissociated figure	weak acid, direct route	$[H^+] = \% \times [\text{acid}]$, then $-\log$
two solutes, the second a sodium salt of the first	buffer	$pH = pK_a + \log\left(\frac{[\text{salt}]}{[\text{acid}]}\right)$
weak base + its salt	alkaline buffer	gives pOH, not pH. Then $14 - pOH$
strong acid + its salt (HCl + NaCl)	not a buffer	treat it as the strong acid alone
weak acid + an unrelated salt	not a buffer	treat it as the weak acid alone
the two concentrations are EQUAL	pH = pK_a exactly	$\log(1) = 0$. Skip the log

PH CONVERSIONS AND STRONG ACIDS		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
the four relationships	$pH = -\log[H^+]$ $[H^+] = 10^{-pH}$ $pOH = -\log[OH^-]$ $[OH^-] = 10^{-pOH}$ pH + pOH = 14	any corner to any other in at most two moves
monoprotic strong acid: HCl, HNO ₃ , HBr	$[H^+] = 1 \times [\text{acid}]$	0.03 M HCl → pH = 1.52
diprotic strong acid: H ₂ SO ₄	$[H^+] = 2 \times [\text{acid}]$	0.015 M → $[H^+] = 0.030$ → pH = 1.52, the same. That is the point of the question
H₂CO₃ , H₂S	diprotic BUT weak	the doubling shortcut is for strong diprotic acids only
strong base	1. $[OH^-] = [\text{base}]$ 2. $pOH = -\log[OH^-]$ 3. pH = 14 – pOH	0.04 M NaOH → pOH 1.40 → pH 12.60
weak acid, % dissociation given	1. $[H^+] = \% \times \text{initial}$ 2. $pH = -\log[H^+]$ 3. if asked, $K_a = \frac{[H^+][A^-]}{[HA]}$	0.010 M, 5% → $[H^+] = 5.0 \times 10^{-4}$ → pH 3.30

PH CONVERSIONS AND STRONG ACIDS		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
the best question in the module	a WEAK acid and a STRONG acid can have the same pH	0.010 M weak and 0.0005 M strong both give 3.30. pH cannot tell you strong from weak – only K _a can

BUFFERS		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
is it even a buffer?	1. is the acid (or base) weak ? strong → no 2. are both partners present? acid alone → no 3. is the salt the conjugate of that same acid ? CH ₃ COOH + NaCl → no	all three must pass
acidic buffer	pH = pK_a + log([salt] ÷ [acid])	salt on top. pK _a = $-\log K_a$
alkaline buffer	$pOH = pK_b + \log\left(\frac{[\text{salt}]}{[\text{base}]}\right)$	gives pOH. Then pH = 14 – pOH
worked — her Q2	HNO ₂ 0.80 M with NaNO ₂ 0.50 M, K _a = 4.0×10 ^{−4}	pK _a = 3.40; pH = 3.40 + log(0.50/0.80) = 3.28
sanity check every time	more salt than acid → ratio > 1 → log positive → pH above pK _a	if it moves the wrong way you inverted the ratio
add a little acid to a buffer	the basic partner eats it	H ⁺ + CH ₃ COO [−] → CH ₃ COOH
add a little base	the acidic partner eats it	CH ₃ COOH + OH [−] → H ₂ O + CH ₃ COO [−]
"buffering capacity"	higher concentration of both, and as near equal as possible	two buffers can share a pH and differ hugely in capacity
"which acid should I use" for a target pH	equal concentrations → pH = pK _a , so pick the acid whose pK _a matches	target 8.6 → the acid with pK _a 8.60
"what ratio" for a target pH	$[\text{salt}] \div [\text{acid}] = 10^{(pH - pK_a)}$	blood: $10^{(7.40 - 6.1)} = \text{about } 20 : 1$

EQUILIBRIUM AND K _c		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
"write the K _c expression"	1. balance the equation first — her deck prints one that is not 2. products on top, reactants underneath 3. each in square brackets, raised to its coefficient 4. multiply, never add	CH ₄ + 2H ₂ S ⇌ CS ₂ + 4H ₂ K _c = $\frac{[CS_2][H_2]^4}{[CH_4][H_2S]^2}$

EQUILIBRIUM AND K _c		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
"value and units"	units = M ^{Δn} , Δn = powers on top – powers below	your worksheet asks for units and no slide teaches them. Δn = 0 → no units
Q _c versus K _c	same algebra	it is Q _c at any moment, and equals K _c only at equilibrium
"essentially only product / reactant?"	K > 10 ² product K < 10 ^{−2} reactant between = comparable	the boundary is 100, not 10. K = 97 is comparable
add reactant / remove product	shifts right	K _c unchanged
increase pressure	shifts to the side with fewer moles of gas	equal moles → no shift. K _c unchanged
increase temperature	shifts to the side that absorbs heat	the only change that moves K_c itself
catalyst	nothing shifts	no shift, no change in K _c , no change in yield. It speeds both directions equally
heat written into the equation	treat heat as a species	... ⇌ products + heat means exothermic, so heating shifts left

ACIDS — WHAT YOUR COURSE DOES NOT TEACH, AND TWO DECK ERRORS
Do not go looking for these. None of them is in A105: ICE tables K _p and partial pressures K _w as a symbol K _a × K _b = K _w the "x is small" approximation pH of a weak base from K _b alone pH of a salt solution leaving pure solids out of K _c . The one real gap: if a question gives a weak acid with only K _a and a concentration and asks for pH, your course has taught no method for it. No worksheet question needs one. If it appears, the general route is $[H^+] = \sqrt{(K_a \times C)}$ — but say where it came from. Two errors in your own slides. One equilibrium slide prints PCl₅ ⇌ PCl₃ + Cl₂ which is impossible — the left side should be PCl ₃ . Another prints SO₂ + O₂ ⇌ 2SO₃ unbalanced; it should be 2SO ₂ . Neither changes an answer, but balance it yourself before writing any K _c . pK_a of ethanoic acid: one slide says 4.76, the rest of your material and your worksheet say 4.74 . Use 4.74 .

HESS'S LAW — COMBINING EQUATIONS		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
“calculate $\Delta_{\text{rxn}}H^\circ$ ”, then “ given that ” and two or three equations	1. write the target equation and box it 2. take each species in the target: is it on the right side? 3. fix each given equation with ONE of the three moves below 4. add them up and cancel what appears on both sides 5. if it does not cancel to the target, a move is wrong	the adjusted ΔH values then add up
species already correct	keep the equation	ΔH unchanged
species is on the wrong side	reverse the equation	flip the sign of ΔH . Size does not change
coefficient is wrong by a factor	multiply (or halve) the equation	multiply ΔH by the same factor . Sign does not change
both wrong at once	reverse and scale — they are independent	Q9: reverse gives +966, then halve gives +483
worked — her Q1, nothing to fix	$\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$ (+180); $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ (–112). NO cancels	$180 + (-112) = +68 \text{ kJ}$
worked — her Q2, two reversals	NH_3 and HCl must both be reactants but are products, so reverse both: +46.1 and +92.3. Keep the third	$46.1 + 92.3 - 314.4 = -176.00 \text{ kJ}$
worked — her Q7, reverse then halve the result	reversing and adding gives $2\text{H}_2\text{O}_2$, but formation is per mole	$(-572 + 196) \div 2 = -188.00 \text{ kJ/mol}$

ENTHALPY FROM TABLES		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
a table of $\Delta_f H^\circ$ values, one per substance	$\Delta_{\text{rxn}}H^\circ = \Sigma n(\text{products}) - \Sigma n(\text{reactants})$	n is the coefficient from the balanced equation
an element appears in the equation	its $\Delta_f H^\circ$ is zero — drop it	the seven diatomics too: H_2 , N_2 , O_2 , F_2 , Cl_2 , Br_2 , I_2
worked — her Q6	$4\text{FeS}_2 + 11\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 + 8\text{SO}_2$	$[2(-824) + 8(-297)] - [4(-178) + 0] = -4024 + 712 = -3312.00 \text{ kJ}$
a free self-check on her own paper	Q2 and Q8 are the same reaction by two different routes	both give -176.00 kJ . If they disagree, one is wrong

ENTHALPY FROM TABLES		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
a table of bond enthalpies instead	$\Delta H = \text{breaking (+)} + \text{forming (–)}$	reactants first here — the opposite way round from the formation route
worked — her Q10, propanone burning	break $6(\text{C–H}) + 2(\text{C–C}) + 1(\text{C=O}) + 4(\text{O=O}) = +5957$ form $6(\text{C=O}) + 6(\text{O–H}) = -7608$	-1651.00 kJ . A combustion that comes out positive is always wrong
bond counting	each CO_2 has two C=O . Each H_2O has two O–H . Each O_2 is one O=O to break	so 3CO_2 gives 6, not 3

ENTROPY		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
“is there an increase or decrease in entropy”, no numbers	1. count the particles: more on the right means ΔS up 2. compare the states: solid < liquid < aqueous < gas 3. state the sign	give BOTH tests — one of her questions has no particle change and only the state test answers it
fast check	count the moles of gas . Gas dominates	gas up $\rightarrow$ positive. Gas down $\rightarrow$ negative
worked — her Q2, the trap	$\text{NaOH(s)} + \text{acid(aq)} \rightarrow \text{salt(aq)} + \text{H}_2\text{O(l)}$. Particles $2 \rightarrow 2$, no change	but solid $\rightarrow$ aqueous , so ΔS is positive
a table of S° values	$\Delta S = \Sigma S(\text{products}) - \Sigma S(\text{reactants})$	elements are NOT zero here . O_2 has $S^\circ = 205.0$
worked — her Q4(b)	$\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$	$2(193.0) - [191.5 + 3(131.0)] = -198.50 \text{ J mol}^{-1}\text{K}^{-1}$
a second self-check	that -198.5 J is the –0.199 kJ she is handed two questions later	same reaction, same worksheet. They must agree

FREE ENERGY AND SPONTANEITY		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
ΔH and ΔS given, and the word spontaneous	1. T into kelvin: $K = ^\circ\text{C} + 273$ 2. check ΔS units — if J, divide by 1000 3. $\Delta G = \Delta H - T\Delta S$ 4. state the sign AND the verdict in words	$\Delta G < 0$ spontaneous > 0 not = 0 equilibrium
the unit trap, live on her own sheet	Q5 gives ΔH in kJ and ΔS in J — convert. Q6 gives ΔS already in kJ — do not	read the unit every time . Two questions apart, deliberately

FREE ENERGY AND SPONTANEITY		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
the double negative	$-T \times (-\Delta S)$ is PLUS	Q6: $-92.2 + 59.30 = -32.90$, not -151.5
worked — her Q5, ice melting	$\Delta H = +6.01 \text{ kJ}$, $\Delta S = 22.0 \text{ J} = \mathbf{0.0220 \text{ kJ}}$, $T = 298 \text{ K}$	$6.01 - (298)(0.0220) = \mathbf{-0.55 \text{ kJ/mol}}$, spontaneous
worked — her Q6, at two temperatures	$\Delta H = -92.2$, $\Delta S = -0.199 \text{ kJ}$	$298 \text{ K} \rightarrow \mathbf{-32.90}$ spontaneous $773 \text{ K} \rightarrow \mathbf{+61.63}$ not
“at what temperature”, “above what T”	set $\Delta G = 0$, so $T = \Delta H \div \Delta S$. Same units both sides	$Q5 \rightarrow 6.01/0.0220 = 273.2 \text{ K}$ — the melting point of ice, recovered
which side of the crossover is spontaneous	both $+$ $\rightarrow$ spontaneous above T both $-$ $\rightarrow$ spontaneous below T	the other two cases never change verdict

WHEN IS IT SPONTANEOUS — THE FOUR CASES		
	ΔH NEGATIVE (EXOTHERMIC)	ΔH POSITIVE (ENDOTHERMIC)
ΔS positive	spontaneous at ALL temperatures	spontaneous at high T , once $T\Delta S$ is big enough
ΔS negative	spontaneous at low T , while $T\Delta S$ is small	NEVER spontaneous
why	only the $T\Delta S$ term changes with temperature	if ΔH and ΔS agree, T cannot change the verdict. If they conflict, T decides
worked — her Q7	$\text{HI(s)} \rightarrow \text{H}^+ + \text{I}^-$. Heat given out so ΔH is negative ; 1 particle $\rightarrow$ 2 and solid $\rightarrow$ aqueous so ΔS is positive	top-left cell $\rightarrow$ spontaneous at every T , so the chemist’s claim is wrong

ENERGY — WHAT TO REMEMBER ABOUT UNITS AND MARKS
Units, her deck's own rule: values you are given from a table are kJ/mol . Values you calculate are kJ — unless the question asks for an enthalpy of formation , which is per mole.
Entropy tables are in J , enthalpies in kJ. They never mix without dividing by 1000.
Signs: $-\Delta H$ exothermic, heat out. $+\Delta H$ endothermic, heat in. Breaking bonds absorbs (+), forming bonds releases (-).
Read "gives out 81.67 kJ" as $\Delta H = -81.67$. The question will not write the minus sign for you.
Your worksheet says: show your workings clearly, and leave answers to 2 decimal places whenever necessary.
A105 does not teach calorimetry. There is no $q = mc\Delta T$ anywhere in your course. If a question looks like it needs one, re-read it.

OXIDATION STATES — THE RULES, IN THE ORDER THEY WIN		
ASK, IN THIS ORDER	ASSIGN	EXAMPLE
is it an uncombined element ?	0 , and stop	S ₈ , O ₂ , P ₄ , Na → all 0
is the whole species one atom with a charge ?	= the charge, and stop	Fe ³⁺ → +3
is it Group 1, Group 2, or F ?	+1, +2, −1 — these never bend	Mg(OH) ₂ → Mg = +2
is it H ?	+1, unless bonded to a metal → −1	NaH, CaH ₂ → H = −1
is it O ?	−2, unless peroxide → −1	H ₂ O ₂ and O ₂ ^{2−} → O = −1
is it Cl, Br or I ?	−1 in a binary compound. If O or F is also there, do NOT assign it — leave it as the unknown	HClO ₄ → Cl is the unknown, not −1
whatever is left	solve: Σ(count × state) = the total charge	neutral → 0 polyatomic ion → its charge
worked	MnO ₄ [−] Mn + 4(−2) = −1	Mn = +7
worked	HClO ₄ (+1) + Cl + 4(−2) = 0	Cl = +7
worked, the neutral-ligand trick	Mn(H ₂ O) ₆ ²⁺ — each H ₂ O is neutral , so contributes 0	Mn = +2
a fast self-check	a state above +7 means you forgot the ion's charge	summing MnO ₄ [−] to 0 gives +8, which is impossible

REDOX — WHO DID WHAT TO WHOM		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
“what is oxidised / reduced”	1. assign oxidation states on both sides 2. find the atoms whose number changed 3. went up → oxidised went down → reduced	nothing changed → not a redox reaction
“which is the oxidising agent”	the oxidising agent is the thing that gets REDUCED	the name says what it does to the other species
self-check on your own sentence	the two words must be opposites	oxidised → reducing agent reduced → oxidising agent. If they match, it is wrong
balance a redox equation, acid	1. split into two half-equations 2. balance everything except O and H 3. add H₂O for O, then H⁺ for H 4. add electrons to balance the charge 5. scale so the electrons cancel, then add	check: same atoms and same total charge each side
balance in base	do all of the above via H⁺ first , then add the same number of OH[−] to BOTH sides and combine into H ₂ O	do not start from OH[−] — it is much harder
“will this metal displace that one”	a metal reduces the compounds of anything below it in the activity series	

CELLS — GALVANIC AND ELECTROLYTIC		
	GALVANIC (VOLTAIC)	ELECTROLYTIC
how to tell in 3 seconds	no battery — just a voltmeter or bulb	there is a power source in the diagram
the reaction	spontaneous	non-spontaneous, must be driven
E° _{cell}	always positive	negative
anode	oxidation, negative	oxidation, positive
cathode	reduction, positive	reduction, negative
what never changes	anode = oxidation, cathode = reduction, in BOTH	electrons flow anode → cathode, in BOTH. Only the sign swaps
the salt bridge	a tube of electrolyte in a gel joining the two half-cells	lets ions flow, stops the solutions mixing
plating or refining	the object you want coated or pure is the cathode	metal always deposits at the cathode

E° AND ELECTROLYSIS		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
which one is reduced, from two E° values	the more positive E° is reduced → it is the cathode	“less positive” counts as more negative. +0.34 vs +0.80 → the +0.34 is oxidised
“calculate E° _{cell} ”	your course writes it as E°_{cell} = E°_{oxidation} + E°_{reduction}	reverse the oxidised half and flip the sign of its E°. Your worksheet's answer table is printed in this form
is it spontaneous?	E° _{cell} positive → yes	negative → no
you multiplied a half-equation	E° is NOT multiplied. Volts are per electron	scale the equation, never the volts. ×2 leaves +0.80 as +0.80
worked	Cr (−0.74) with Ni (−0.28)	Cr oxidised: E° _{ox} = +0.74 E° _{cell} = 0.74 − 0.28 = +0.46 V
electrolysis: mass from current and time	1. Q = I × t, t in seconds 2. moles of electrons = Q ÷ 96500 3. divide by the electrons in the half-equation 4. multiply by the molar mass	2 A for 1 h → Q = 7200 → 0.0746 mol e [−] Na ⁺ + e [−] → Na is 1:1 → 1.72 g
run it backwards for “how long”	moles → ×96500 → Q → ÷ I → seconds → ÷3600 for hours	2.5 mol e [−] at 15 A → 4.47 hours
if the question prints an E° value	use the question's value, not the table's	your table and one worksheet disagree on Ni (−0.25 vs −0.28)

KINETICS		
THE QUESTION SAYS	WHAT TO DO	WORKED, FROM HER PAPER
“why does X speed it up”	every reason ends the same way: more successful collisions	surface area, concentration, pressure → more frequent collisions
temperature	the only factor that does both	more frequent collisions and a bigger proportion above E _a
a catalyst	an alternative route with lower E_a	more molecules clear the lower bar. It is not used up, and k increases
a Maxwell-Boltzmann figure	molecules that can react = the area under the curve to the RIGHT of E_a	shade it, then compare the shaded areas
what moves on that figure	temperature shifts the curve right concentration makes it taller, same place catalyst moves the E _a line, not the curve	three different changes. Do not blur them
“rate = k[A] ^m [B] ⁿ ”	the orders come from experiment , never from the coefficients	one exception: in an elementary step of a mechanism, order = coefficient
find the order from a rates table	find two runs where only one concentration changed. Rate unchanged → zero order doubles → first quadruples → second	
find the order from half-life	t½ constant → first order t½ halves each time → zero t½ doubles each time → second	constant 12 s each halving → first order
units of k in 3 seconds	M ^(1 − overall order) s ^{−1}	overall 1 → s ^{−1} overall 3 → M ^{−2} s ^{−1}
Arrhenius, two conditions	k = Ae ^{−E_a/RT} , so with A constant ln(k ₂ /k ₁) = E _{a1} /RT ₁ − E _{a2} /RT ₂	°C → K first. 400°C is 673 K. R is given in kJ, so E _a in kJ goes straight in

ELECTRONS — GAPS AND ERRATA IN YOUR OWN NOTES	
Use the potential table in the attached Word document , not the short one printed on the slides. They come from different sources and disagree — the attached one is the one your worksheet answers match.	
Not stated anywhere in your notes , so look them up if asked: that a catalyst leaves ΔH, the products and the equilibrium position unchanged ; that the salt bridge maintains electrical neutrality and completes the circuit (your deck says only that it lets ions through and stops mixing); what happens when E° _{cell} is exactly zero ; and cell notation such as Zn Zn ²⁺ Cu ²⁺ Cu, which your course never uses.	
Have a periodic table to hand — one electrolysis question needs the molar masses of Hg and Cl and supplies neither.	
Typos in your slides: the activity series prints Mg twice where the second should be Mn ; a car-battery equation prints PbSO ₂ for PbSO₄ ; one hydrogen product is labelled (s) instead of (g) ; and the iron potential is shown as 0.44 V when it is −0.44 V .	